

Imbalanced Data in Machine Learning – Undersampling, Oversampling, SMOTE

1 What is Imbalanced Data?

- A dataset where **class distribution is skewed**: one (majority) class has far more samples than the other (minority).
- Example: Fraud detection (99% genuine, 1% fraud).

2 Why It's a Problem

- Models trained on imbalanced data tend to be **biased toward the majority class**.
- Metrics like accuracy can be **misleading** (e.g., predicting “no fraud” 99% of the time gives 99% accuracy but 0% fraud detection).
- Leads to:
 1. Poor recall on the minority class.
 2. Overconfidence in majority predictions.

3 Metrics for Evaluation (instead of accuracy)

- **Precision, Recall, F1-Score** (esp. Recall for minority class).
- **ROC-AUC / PR-AUC curves**.
- **Balanced accuracy** = (Sensitivity + Specificity)/2.
- **Cohen's Kappa, Matthews Correlation Coefficient (MCC)**.

4 Causes of Imbalance

- Natural rarity (e.g., rare diseases).
- Data collection bias (e.g., easier to collect negative samples).
- Labeling errors.

5 Strategies to Handle Imbalance

A. Data-Level Methods (Resampling)

Method	Idea	Pros	Cons
Random Undersampling	Remove some majority samples to balance classes.	Simple, fast.	May discard useful data → underfitting.
Random Oversampling	Duplicate minority samples.	Preserves info.	Risk of overfitting (copies).

SMOTE (Synthetic Minority Over-sampling Technique)	Generate synthetic samples for minority using k-nearest neighbors interpolation.	Reduces overfitting vs. simple oversampling.	May create ambiguous points if classes overlap.
Variants of SMOTE	Borderline-SMOTE, ADASYN (focus near class boundary).	More robust in complex spaces.	Slightly more complex.
Cluster-based Over/Under-sampling	Cluster data first, then sample.	Keeps structure.	Needs tuning.

B. Algorithm-Level Methods

- **Class Weighting / Cost-Sensitive Learning**
 1. Assign higher penalty (weight) to minority class misclassifications.
 2. Many algorithms support `class_weight="balanced"` (e.g., Logistic Regression, SVM, Decision Trees).
- **Threshold moving**
 1. Adjust decision threshold instead of default 0.5 to favor minority recall.
- **Anomaly Detection Models**
 1. Treat minority as “outliers.”

C. Ensemble Methods

Approach	Explanation
Balanced Random Forest	Uses undersampled data at each tree bootstrap.
EasyEnsemble	Creates several undersampled subsets of majority + full minority, trains multiple classifiers, aggregates.
Bagging / Boosting with reweighting	E.g., AdaBoost, Gradient Boosting with sample weights.
Balanced XGBoost / LightGBM	Use <code>scale_pos_weight</code> or similar parameter.

D. Other Advanced Techniques

- **Hybrid Sampling** – Combine over & under sampling (e.g., SMOTE + Tomek Links / Edited Nearest Neighbors to clean noise).
- **One-class learning** – Learn only from majority and detect deviations.

- **Data augmentation (domain-specific)** – e.g., rotate/flip images of minority class.
 - **Active learning** – Focus labeling effort on rare cases.
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6 Practical Workflow

1. **Explore data** → Identify imbalance ratio (IR = majority/minority).
 2. **Pick metric** that reflects business goal (recall, F1, etc.).
 3. **Try simple solutions first:**
 1. Class weights.
 2. Random under/over sampling.
 4. **Use SMOTE or advanced samplers** for non-linear boundaries.
 5. **Consider ensembles** if data is large.
 6. **Validate carefully:**
 1. Use stratified k-fold cross-validation.
 2. Ensure resampling only on training folds (avoid leakage).
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7 Tips & Best Practices

- Resampling is powerful but can distort distribution — always visualize results.
 - Use **probability calibration** if resampling affects predicted probabilities.
 - When dataset is huge, undersampling may be enough.
 - When minority is extremely tiny (<1%), anomaly detection or cost-sensitive learning can be better than heavy oversampling.
 - For multiclass imbalance, consider one-vs-all weighting or resampling per class.
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Quick Recap

Imbalanced data skews learning → focus on metrics, resampling, algorithm tweaks, or ensembles.

Undersampling reduces majority, **Oversampling** adds minority, **SMOTE** synthesizes new data, and **Ensemble/Cost-sensitive** methods further improve fairness.
